JAVA PROGRAMMING

Java is a high-level, object-oriented and platform-independent programming language developed by Sun Microsystems in 1995.

Java follows the principle of "Write Once, Run Anywhere" because Java programs are compiled into bytecode which can run on any system having a Java Virtual Machine (JVM).

Features of Java:

1. Simple
2. Object-Oriented
3. Platform Independent
4. Secure
5. Robust
6. Portable
7. Multithreaded
8. High Performance

Java is widely used in Web Applications, Enterprise Applications, Mobile Applications, Desktop Applications and Cloud Computing.

JVM, JRE AND JDK

JVM stands for Java Virtual Machine.

Responsibilities of JVM:

1. Loads class files
2. Verifies bytecode
3. Executes bytecode
4. Manages memory

JRE stands for Java Runtime Environment.

JRE contains:

1. JVM
2. Libraries
3. Supporting files

JDK stands for Java Development Kit.

JDK contains:

1. JRE
2. Compiler (javac)
3. Debugging tools
4. Development utilities

Relationship:

JDK > JRE > JVM

VARIABLES AND DATA TYPES

Variables are used to store data.

Syntax:

```
datatype variableName;
```

Example:

```
int age = 25;
```

Primitive Data Types:

1. byte
2. short
3. int
4. long
5. float
6. double
7. char
8. boolean

Reference Types:

1. String
2. Arrays
3. Classes
4. Interfaces

Example:

```
String name = "Syed Abdul Rehman";
```

OBJECT ORIENTED PROGRAMMING

The four pillars of OOP are:

1. Encapsulation

Binding data and methods together.

2. Inheritance

Acquiring properties from parent class.

3. Polymorphism

One interface, multiple forms.

4. Abstraction

Hiding implementation details.

Benefits:

1. Code Reusability
2. Security
3. Maintainability
4. Scalability

INHERITANCE

Inheritance allows one class to acquire properties and methods of another class.

Keyword:

extends

Example:

```
class Animal {
  void eat() {}
}

class Dog extends Animal {
  void bark() {}
}
```

Types of Inheritance:

1. Single
2. Multilevel
3. Hierarchical

Advantages:

1. Code Reusability
2. Reduced Redundancy
3. Easy Maintenance

POLYMORPHISM

Polymorphism means many forms.

Types:

1. Compile Time Polymorphism

Method Overloading

Example:

```
add(int a, int b)
add(int a, int b, int c)
```

2. Runtime Polymorphism

Method Overriding

Example:

```
Parent reference = new Child();
```

Advantages:

1. Flexibility
2. Reusability
3. Extensibility

EXCEPTION HANDLING

Exception Handling is used to handle runtime errors.

Keywords:

1. try
2. catch
3. finally
4. throw
5. throws

Example:

```
try {
  int x = 10 / 0;
}
catch(Exception e) {
  System.out.println(e);
}
```

Advantages:

1. Prevents Program Crash
2. Improves Reliability
3. Improves Maintainability

COLLECTION FRAMEWORK

The Collection Framework provides ready-made data structures.

Interfaces:

1. List
2. Set
3. Queue
4. Map

Implementations:

List:

1. ArrayList
2. LinkedList

Set:

1. HashSet
2. TreeSet

Map:

1. HashMap
2. TreeMap

Benefits:

1. Dynamic Storage
2. Faster Searching
3. Easier Data Management

MULTITHREADING

Multithreading allows multiple tasks to execute simultaneously.

Ways to create thread:

1. Extend Thread class

```
class MyThread extends Thread {
    public void run() {
        System.out.println("Thread Running");
    }
}
```

2. Implement Runnable Interface

```
class MyRunnable implements Runnable {
    public void run() {
        System.out.println("Running");
    }
}
```

Advantages:

1. Better Performance
2. Resource Sharing
3. Concurrent Execution

JDBC

JDBC stands for Java Database Connectivity.

JDBC is used to connect Java applications with databases.

Steps:

1. Load Driver
2. Establish Connection
3. Create Statement
4. Execute Query
5. Process Result
6. Close Connection

Important Interfaces:

1. DriverManager
2. Connection
3. Statement
4. PreparedStatement
5. ResultSet

Advantages:

1. Database Connectivity
2. Dynamic Queries
3. Data Persistence